

Received 7 October 2024, accepted 18 November 2024, date of publication 27 November 2024,
date of current version 16 December 2024.

Digital Object Identifier 10.1109/ACCESS.2024.3506973

RESEARCH ARTICLE

Anomaly Detection of Deepfake Audio Based on Real Audio Using Generative Adversarial Network Model

DAEUN SONG^{ID}, NAYOUNG LEE^{ID}, JIWON KIM^{ID}, AND EUNJUNG CHOI^{ID}

Department of Information Security, Seoul Women's University, Seoul 01797, South Korea

Corresponding author: Eunjung Choi (chej@swu.ac.kr)

This work was supported in part by Seoul Women's University under Grant 2024-0182; and in part by Korean Government [Ministry of Education (MOE)] [2024 Government Collaboration-Type Training Project (Information Security Field), No. 2024 Personal Information Protection-002].

ABSTRACT Deepfake audio causes damage not only to individuals and companies, but also to nations; therefore, research on deepfake audio detection technology is crucial. Most existing deepfake audio detection research has been conducted using supervised learning; however, when a new synthetic deepfake audio emerges, real-time detection becomes difficult because of the limitations of supervised learning. Therefore, this paper proposes a new anomaly detection technique for identifying deep-fake audio using unsupervised learning. This method involves learning the feature distribution of numerous real human voices and then calculating an anomaly score for each voice to determine whether it is deepfake. In this study, we imaged speech using mel-spectrogram and mel-frequency cepstral coefficient (MFCC), which are speech preprocessing methods. Subsequently, the parameters of the GANomaly and f-AnoGAN models, which are effective in detecting abnormalities in speech, were tuned and subjected to unsupervised training. The most effective result had an F1-score of 0.93 in and was obtained by combining imaging speech with Mel-Spectrogram with training using the GANomaly model.

INDEX TERMS Anomaly detection, Deepfake, Deepfake audio, deep learning, f-AnoGAN, feature extraction, GANomaly, generative adversarial network model, mel-spectrogram, MFCC, unsupervised learning.

I. INTRODUCTION

In January 2024, prior to the US presidential election, an attempt was made to influence the voters in New Hampshire, urging them not to vote in the primary election using the voice of the presidential candidate Joe Biden [1], [2], [3]. In March 2022, a video of Ukraine's president was released during the Russia-Ukraine War. In the video, people were told to surrender and abandon the war, which was later revealed to be a deep fake audio [4]. The impact of deepfake audios extends from individuals and businesses to an entire country. Its emergence not only enhances the authenticity of existing deepfake videos but also has the potential to mislead when used independently. To prevent cyber criminals from utilizing deepfake audio for impersonation, fraud, and voice phishing, further research is required on detecting deepfake audios.

The associate editor coordinating the review of this manuscript and approving it for publication was Cesar Vargas-Rosales^{ID}.

Most of the current research on deepfake audio detection utilizes supervised learning techniques. In [5], a supervised learning algorithm, support vector machine (SVM), was used for classification, using mel frequency cepstral coefficient (MFCC) features extracted from FoR datasets to identify deepfake audios. In [6], a convolutional neural network–long short-term memory (CNN-LSTM) supervised learning model and MFCC features from the ASVSpooF-2019 dataset were used to effectively detect deep fake audio. Other studies have also utilized CNN as supervised learning models [7], [8], [9].

Supervised-learning-based deepfake audio detection demonstrates excellent performance on training data through extensive training using labeled deepfake audio data. However, accuracy is not guaranteed for untrained datasets. Therefore, additional training data are required as new types of deepfake audios emerge [10]. Meanwhile, deepfake audio is evolving, being created in a variety of ways with traces of synthesis removed [11], which makes it difficult to collect

and train on all types of audio depending on how they were created.

To address this problem, this study aims to detect deep fake audio using anomaly detection based on an unsupervised model. Anomaly detection involves identifying patterns that do not match expected behavior [12]. We trained the anomaly detection system using unlabeled real human speech to identify audio that deviates from the pattern of real audio as deepfake audio. This approach allows the detection of new deepfake audio without requiring additional data training.

The generative adversarial network (GAN) model offers the advantage of being able to train on unlabeled data. Furthermore, their capacity to learn complex patterns and distributions within the data enables them to outperform other models in detecting anomalous audio. [13]. The GAN-based anomaly detection model, GANomaly, is well known for its abnormal detection capabilities, achieving an area under the receiver operating characteristic (AUC) of 0.9822 in recognizing abnormal sounds in motors by detecting anomalies in untrained data [14]. Additionally, the f-AnoGAN model demonstrated an AUC of 0.9301 for image-level anomaly detection, outperforming other models in terms of accuracy [15]. Therefore, we performed unsupervised learning using the GANomaly and f-AnoGAN models.

To perform unsupervised learning, we utilized the mel spectrogram and MFCC to train the model by visualizing normal audio features as the basis for anomaly detection. The mel-spectrogram represents human speech frequencies well [16] and has been applied to speech recognition across different fields [17], [18]. The MFCC is a coefficient that quantifies the frequency features extracted from speech signals [19] and is predominantly used in current deepfake audio detection research.

In this study, we used the mel-spectrogram and MFCC preprocessing techniques to extract the features of normal audio, with GANomaly and f-AnoGAN models used for unsupervised learning. The goal was to propose the most effective method for detecting deep fake audio.

The remainder of this paper is organized as follows. The Related Work section discusses audio datasets, features, and research using unsupervised models. The Methodology section outlines the overall framework of the experiments, including datasets, preprocessing models, and metrics used. The Experiments section details the experimental process and results, and the Conclusion summarizes our findings.

II. RELATED WORK

A. DATASET

The datasets examined in this paper are listed in Table 1. The Record category includes imported voice datasets, while Video Extraction refers to datasets that specifically utilize celebrity voices. Data that do not fall under the categories of actual recorded human voices or video extraction are classified as Synthesis. A detailed description of the dataset follows below.

ASVSpooof 2019 considers logical access (LA) and physical access (PA) scenarios in three main forms of spoofing attacks: synthesized, transformed, and replayed speech. The training data include speech samples from 20 speakers (8 male, 12 female); the development data include 10 speakers (4 male, 6 female); and the evaluation data include samples from 40 speakers (21 male, 27 female). The training and development sets contain spoofed attacks generated from known attacks, while the evaluation set contains attacks generated from unknown attacks [29], [30].

ASVSpooof 2021 comprises logical access (LA), physical access (PA), and voice-deep fake (DF) datasets. LA is an extended version of the ASVSpooof 2015 data, whereas PA is an extension of the ASVSpooof 2019 data. The DF dataset is new, reflecting the advances in deep-fake audio generation techniques, with scenarios covering the compromised voices of celebrities or ordinary individuals. The training and development sets contain known attacks, whereas the evaluation set contains unknown attacks [31], [32].

WaveFake includes ten sample datasets generated by six different network architectures. It contains 196 hours of audio and 117,985 generated audio clips (16-bit PCM wav) in English and Japanese [26].

Fake or Real dataset consists of over 195,000 speech samples, which including authentic and deep-learning-synthesized speech. These samples were generated from open-source speech datasets and real-world speech sources, such as TED Talks and Youtube videos [33].

M-AILABS Speech is available in eight languages: English, French, German, Italian, Polish, Russian, Spanish, and Ukrainian. Based on LibriVox and Project Gutenberg, it contains approximately 1000 h of audio with 493,900 basic human utterances sampled at 16 kHz [34].

ADD 2022 dataset was developed to address the gap in the ASVSpooof 2021 dataset, which contains DFs with different background noise and distractions, DF clips generated by combining real speech and deep fake audio, and DFs generated using a new algorithm [35], [36].

B. PRE-PROCESSING

The speech preprocessing methods investigated in this study are listed in Table 2 and described in detail below.

Mel-spectrogram compresses frequency using the mel-scale, which represents the difference between sounds based on human hearing in a logarithmic expression [44].

Mel frequency cepstral coefficients are obtained by taking the cosine transform of the natural logarithm of short-term energy spectrum represented in the mel-frequency scale [19]. This method is advantageous as it requires less complex feature extraction algorithms for implementation [45]. The MFCC is commonly employed for deepfake audio detection.

Bispectrum is a type of frequency analysis that detects the presence of components resulting from the interaction between two or more frequencies. It is primarily used in signal processing and medicine [42].

TABLE 1. Organizing datasets available for deepfake audio research.

Dataset	Num of Data	Languages	File Type	Utterances		Creation Method			Ref	Src	Year
				Real	Fake	Record	Video Extraction	Synthesis			
ASV Spoof 2019	121,461	ENG	N/A	O	O	O	X	O	[6, 20, 21, 22, 23, 24, 25]	ASV spoof	2019
ASV Spoof 2021	1,566,000	ENG	FLAC	O	O	O	X	O	[22, 23, 25]	ASV spoof	2021
WaveFake	117,985	ENG, JPN	WAV	O	O	O	X	O	[24, 25]	[26]	2021
FoR (Fake or Real)	195,000	ENG	WAV	O	O	O	O	O	[5, 25]	York University	2019
M-AILABS Speech	493,900	ENG, FRA, GER, ITA, POL, RUS, SPA, UKR	WAV	O	X	O	X	X	[27]	M-AILABS	2019
ADD 2022	88,035	CHI	N/A	O	O	O	X	O	[28]	ADD Challenge	2022

TABLE 2. Pre-processing to convert speech to images.

Method	Function	Reference	Source
Mel Spectrogram	$Mel(f) = 2595 \log \left(1 + \frac{f}{700} \right)$	[37, 38]	-
MFCC	$MFCC(d) = \sum_{k=1}^K X_k \cos \left[d(k - 0.5) \frac{\pi}{K} \right]$	[5, 6, 39]	[19]
Bispectrum	$B(f_1, f_2) = F(f_1) \cdot F(f_2) \cdot F^*(f_1 + f_2)$	[40, 41]	[42]
Bicoherence	$B_c(\omega_1, \omega_2) = \frac{Y(\omega_1)Y(\omega_2)Y^*(\omega_1 + \omega_2)}{\sqrt{ Y(\omega_1)Y(\omega_2) ^2} \sqrt{ Y(\omega_1 + \omega_2) ^2}}$	[37]	[43]

Bicoherence is a forensic technique used to differentiate human and synthetic speech. The current speech synthesis algorithms exhibit unusually high-order bispectral correlations not found in human speech [43].

C. MODEL

Unsupervised methods have been used in various audio and anomaly detection studies, including audio anomaly detection. These studies using unsupervised models are presented in Table 3. Datasets collected directly from research or unknown datasets are marked with asterisks. The highest accuracy was obtained when multiple conditions were tested within a study. A detailed description follows.

In audio research and anomaly detection, the GANomaly model is utilized [56]. In [14], faults were detected based on the sound of an induction motor using GANomaly, achieving an accuracy of 92.52%. In [47], an audio anomaly-detection technique was proposed to detect state abnormalities based

on the sound of a machine using a GANomaly model. In [48], the proposed hybrid model combines GANomaly and NSGA-III to detect anomalies in cloud environments. In [49], an anomaly detection method using GANomaly is proposed to analyze and detect abnormal behaviors in real-time.

Audio research using the f-AnoGAN model has also been conducted. In [50], an anomaly-detection method using the f-AnoGAN model was proposed to identify defects in smart processes. In [51], the use of an f-AnoGAN model was proposed for anomaly detection, to identify defects in manufacturing processes. An accuracy of 86% was achieved. In [52], a study on anomaly detection was conducted, using an f-AnoGAN to address the problem of a small number of samples in belt defects, achieving an accuracy of 98.13%.

Audio research has also been conducted using unsupervised learning methods other than GANomaly and f-AnoGAN. In [53], a study was conducted on unsupervised anomaly detection using autoencoders and Neyman-Pearson

TABLE 3. Study of unsupervised models used in audio research or anomaly detection.

Model		Src	Ref	Audio research	Anomaly research	Dataset	Preprocessing	Performance Metrics		ACC	year
GANomaly		[46]	[14]	O	O	*	Mel-spectrogram	AUROC	0.98	92.52%	2019
			[47]	O	O	DCASE 2020 Task 2	Log-Mel spectrogram	N/A	N/A	N/A	2021
			[48]	X	O	CSE-CIC-IDS2018 dataset	N/A	N/A	N/A	98.87%	2022
			[49]	X	O	N/A	N/A	N/A	N/A	N/A	2022
f-AnoGAN		[15]	[50]	X	O	*	N/A	N/A	N/A	N/A	2022
			[51]	X	O	MVTec anomaly detection dataset (private)	N/A	AUC	0.85	86%	2023
			[52]	X	O	*	N/A	AUC	0.99	98.13%	2023
etc	Autoencoder	-	[53]	O	O	Fisher English database	MFCC	AUC	N/A	N/A	2018
	Isolation Forest	-	[54]	O	O	*	N/A	F1	0.56	N/A	2018
	t-SNE	-	[55]	O	X	*	Spectrogram	N/A	N/A	N/A	2020

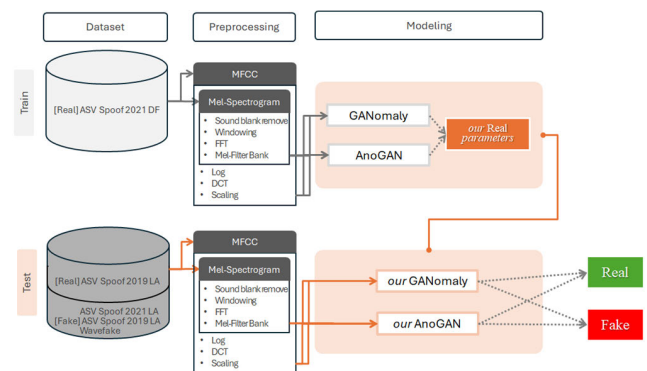
theorem to detect anomalous sounds. In [54], an isolation forest model was used to detect audio anomalies in wireless transmissions through a modular IoT gateway. In [55], the t-distributed stochastic neighbor embedding (t-SNE) model was used to obtain sound data of interest with high precision.

III. METHODOLOGY

A. FRAMEWORK

The proposed framework comprises training and testing phases, as shown in the Fig. 1. We first preprocess using the Mel-spectrogram and MFCC, and then perform GANomaly and f-AnoGAN modeling. During the training step, the real audio of the ASV Spoof 2021 DF was processed using Mel-Spectrogram and MFCC, which was converted into an image from which the audio features were extracted through preprocessing. Subsequently, two anomaly detection models, GANomaly and f-AnoGAN, were trained using the preprocessed results. The optimal model for detecting deepfake audio was determined through hyperparameter-tuning during the unsupervised learning process.

During the testing phase, the same preprocessing method was applied to real and deepfake audios from ASVspoof 2019 LA, deepfake audios from ASVspoof 2021 DF, and audios from the WaveFake dataset. They were then input into the proposed model to obtain the anomaly score. In the process, since only the features of real audio are learned, the difference in anomaly scores between real and deepfake audio becomes clearly noticeable, which can improve the performance of real and deepfake audio classification.

**FIGURE 1.** Frameworks.

B. DATASET

In this study, a real audio from the ASVspoof 2021 DF [31] was used as training data because we used an anomaly detection model based on unsupervised learning with anomaly detection criterion set to real audio. The test dataset included real and deepfake audios from ASVspoof 2019 LA [29], deepfake audio from ASVspoof 2021 DF, and the WaveFake dataset [26].

The ASV spoof dataset is the most widely used dataset for deepfake audio detection (ASVspoof 2019 in [57], [58], [59] and ASVspoof 2021 in [23], [60], [61]). Recent deep fake audio-detection studies combine these two datasets for training and evaluation [62], [63], achieving outstanding results. The WaveFake dataset generates speech deepfakes using different neural network architectures and has been actively used in deepfake audio detection studies [5], [24], [64]. Therefore, we adopted this dataset in this work.

TABLE 4. Preprocessing parameters.

	n_fft	hop_length	n_mels
Case 1(Mel-spectrogram)	2048	512	128
	2048	512	40
	2048	512	20
Case 2(MFCC)	2048	512	128
	2048	512	40
	2048	512	20

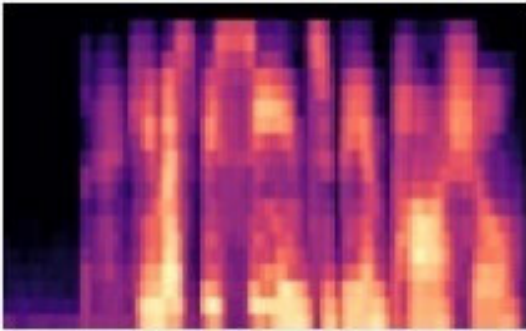


FIGURE 2. Mel20 of real audio.

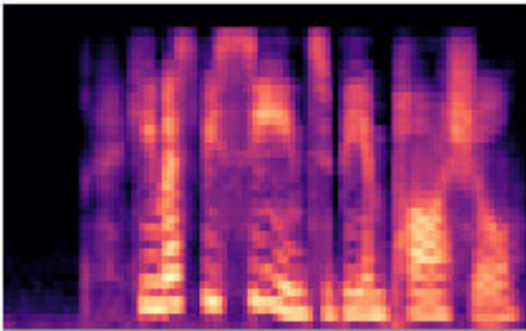


FIGURE 3. Mel40 of real audio.

C. PREPROCESSING

For preprocessing the audio, we used the Mel-spectrogram and MFCC. In the traditional analysis, spectrograms [65] were used to examine audio, with the mel-spectrogram being particularly useful for analyzing human speech This technique emphasizes the low-frequency spectrum, where most of the energy is concentrated using a bank of mel filters. MFCC is a process for obtaining a Mel spectrogram by taking the logarithm and then performing a discrete cosine transform (DCT). We adopted mel-spectrogram and MFCC as they are widely used in existing deepfake audio detection research [6], [39], [66].

1) MEL-SPECTROGRAM

The mel-spectrogram is a visual representation of the changing frequency spectrum of an audio signal and considers both the time and frequency of the signal. The process of creating a spectrogram consists of decomposing the audio using the short-time Fourier transform (STFT), to break the signal into small bins, thereby analyzing the frequencies in each of those bins to see how they change over time. However, distinguish-

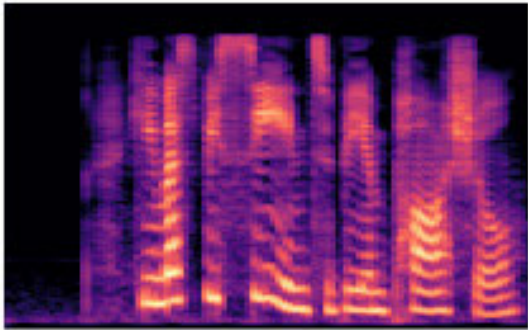


FIGURE 4. Mel128 of real audio.

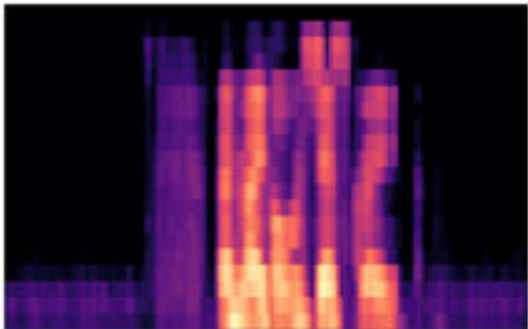


FIGURE 5. Mel20 of fake audio.

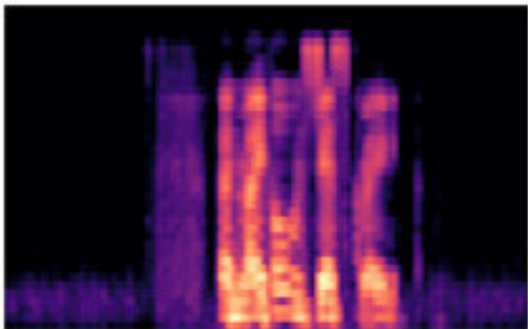


FIGURE 6. Mel40 of fake audio.

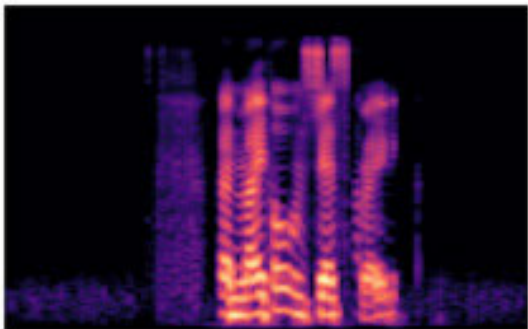


FIGURE 7. Mel128 of fake audio.

ing differences at high frequencies is more challenging for human hearing than at low frequencies [67]. For example, distinguishing between 500 Hz and 1000 Hz is much easier than between 10000 Hz and 10500 Hz. Thus, the mel-spectrogram was introduced to distinguish the differences at these high frequencies.

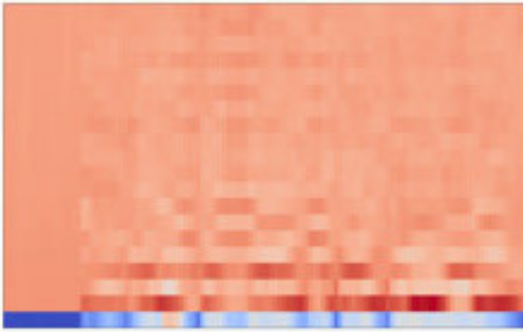


FIGURE 8. MFCC20 of real audio.

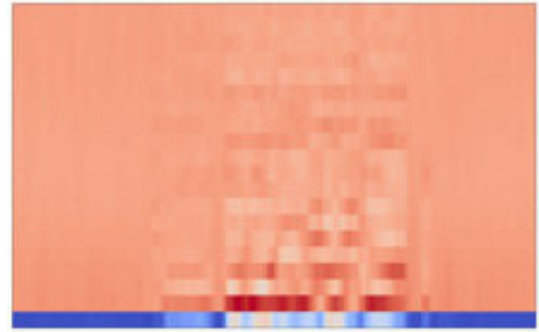


FIGURE 11. MFCC20 of fake audio.

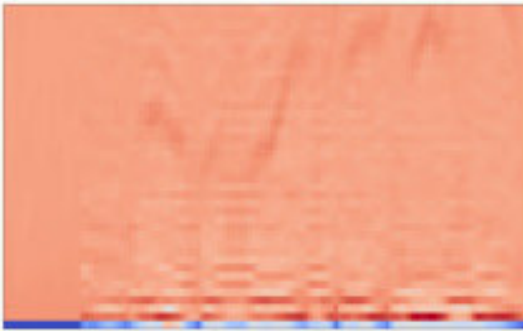


FIGURE 9. MFCC40 of real audio.

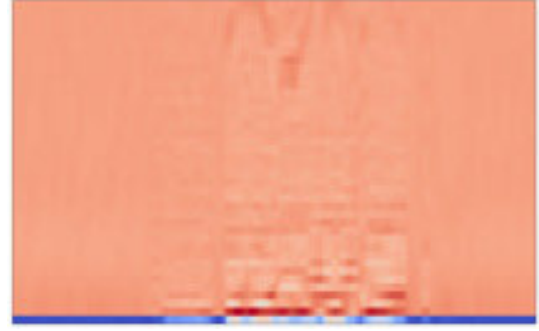


FIGURE 12. MFCC40 of fake audio.



FIGURE 10. MFCC128 of real audio.

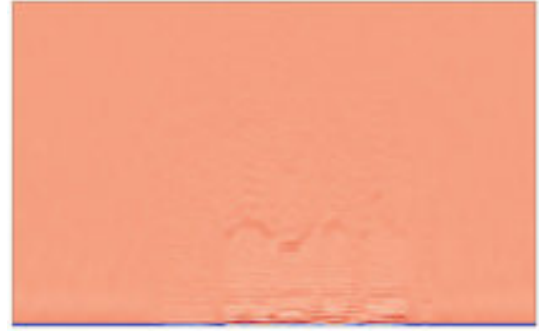


FIGURE 13. MFCC128 of fake audio.

The mel-spectrogram is a pre-processing technique used to provide a sound information model similar to that perceived by humans [44]. To create a mel-spectrogram, a mel filter bank is applied to the spectrogram to convert the frequency to a mel scale. The mel filter bank is based on how humans perceive sound and convert frequencies linearly up to 1 kHz and then logarithmically above that [68].

In this work, we used the librosa library [69] to preprocess the audio, first by removing silent sections from the audio data and then computing the STFT with window size (n_{fft}) set to 2048 and overlap between windows (hop_length) set to 512. The window size (n_{fft}) is used when necessary to convert a continuous signal into a frequency with higher frequency resolution. The overlap size (hop_length) value between windows refers to the spacing between them when cutting the audio into smaller pieces [70], thereby setting the mel filter (n_{mels}) value to create a mel filter bank. Higher mel-filter values for n_{mels} provide more detailed

information but may require more computational resources and memory usage.

Therefore, we conducted experiments to determine the optimal value of the mel filter for deepfake audio detection, considering both performance and cost. The default mel filter value in the official librosa documentation is 128 and ranges from 20 to 40 in speech recognition [19], [71], [72]. The pre-processing parameters for each case are detailed in Table 4, and the mel-spectrograms for real and deepfake audio are shown in Figs. 2-7.

2) MFCC

In MFCC, a feature is extracted by taking the logarithm of the mel-spectrogram and applying DCT. The MFCC of the real audio is shown in Figs. 8-13. However, normalization is particularly beneficial when the distribution of features is too low, as illustrated in Figs. 8-13, indicating that the values are very small or within a limited range. Therefore, scaling

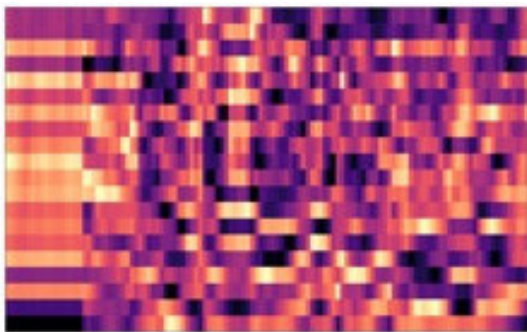


FIGURE 14. Scaled MFCC20 of real audio.

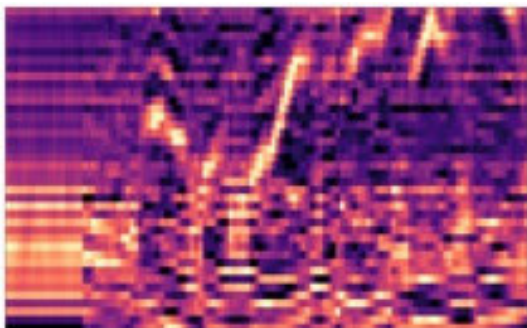


FIGURE 15. Scaled MFCC40 of real audio.

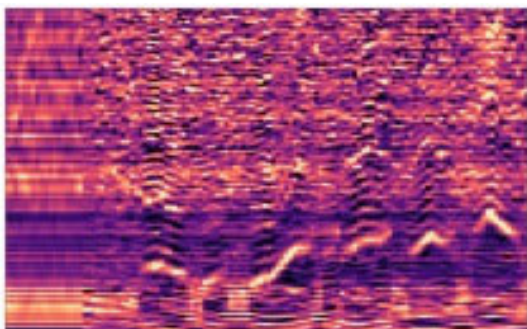


FIGURE 16. Scaled MFCC128 of real audio.

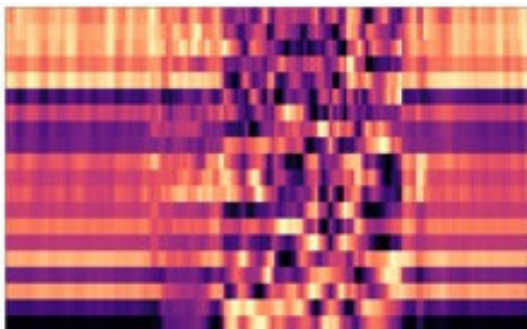


FIGURE 17. Scaled MFCC20 of fake audio.

is performed to generate a normalized MFCC in Case 2. The scaled MFCC is shown in Figs. 14-19.

D. MODEL

In the experiments, we selected GANomaly and f-AnoGAN as the most effective unsupervised anomaly detection models.

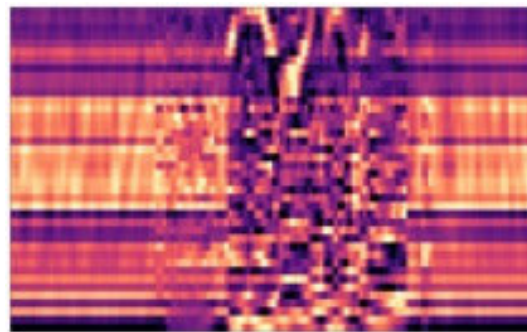


FIGURE 18. Scaled MFCC40 of fake audio.

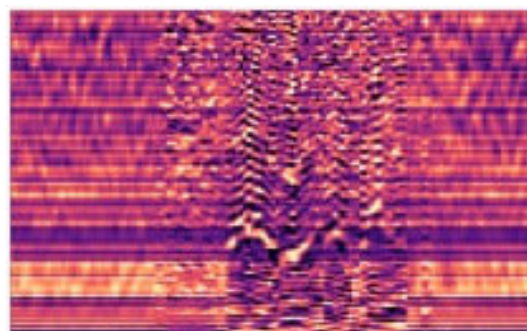


FIGURE 19. Scaled MFCC128 of fake audio.

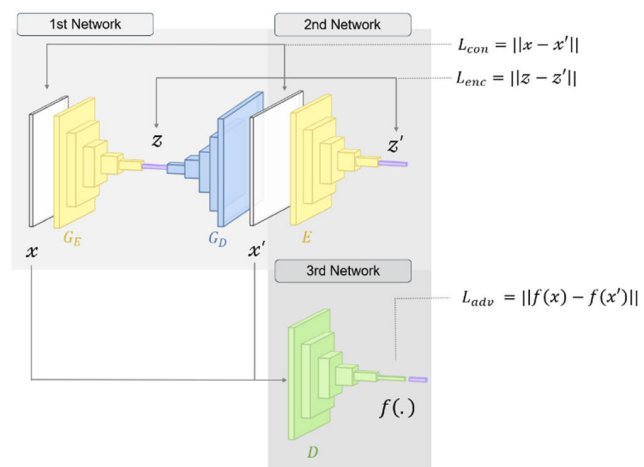


FIGURE 20. GANomaly model framework.

1) GANOMALY

The GANomaly [46] model is a variation of GAN [73] and is effective for unsupervised anomaly detection. This model's advantage compared to traditional autoencoders (AEs) is enhancing the decoder's ability to reconstruct input data through the interaction between the generator and discriminator. By combining the reconstruction ability of AE and generation ability of adversarial learning, it can effectively extract the features of the original data from the deep latent space, making it resistant to noise interference.

The GANomaly model has three sub-networks, as shown in Fig. 20. The first network in Fig. 20 serves as the generator

G. It learns the features of the input data x and reconstructs x using the encoder G_e and decoder G_d networks. The generator G takes the input data x and sends it to the encoder network G_e . During this process, the convolutional layer is used to downscale the input data x and batch-normalization and leaky ReLU() activation are performed to compress it into a vector z . The decoder G_d is based on the DCGAN generator [69] architecture, utilizing convolutional transpose layers, ReLU() activation, and batch-norm to upscale vector z and reconstruct the input data x as x' . The second network in Fig. 20 is the encoder network E . This network compresses the image x' reconstructed by the generator G . The encoder E scales down x' to obtain vector z' , represented as $z' = E(x')$. The third network in Fig. 20 is the discriminator network D . Its function is to classify the input data x and output data x' as either real or fake. This network utilizes the standard discriminator network introduced in DCGAN [74]. The GANomaly model combines three loss functions for detection optimization. A loss function is a metric used to evaluate model performance by measuring the discrepancy between the model's predictions and the true, ground-truth values. Adversarial loss aims to minimize the distance between $f(x)$ and $f(x')$; $f(x)$ is the output of the discriminator D for input data x , and $f(x')$ is the output for reconstructed data x :

$$L_{adv} = E_{x \sim p_X} \|f(x) - E_{x \sim p_X} f(G(x))\|_2 \quad (1)$$

Contextual loss minimizes the distance between the input data x and reconstructed image x' .

$$L_{con} = E_{x \sim p_X} \|x - G(x)\|_1 \quad (2)$$

Encoder loss minimizes the distance between the latent feature vectors z and z' generated by encoder E as follows:

$$L_{enc} = E_{x \sim p_X} \|G_E(x) - E(G(x))\|_2 \quad (3)$$

The GANomaly loss function is calculated by combining these three functions, and the significance of each function is adjusted by changing weight w .

$$L = W_{adv}L_{adv} + W_{con}L_{con} + W_{enc}L_{enc} \quad (4)$$

During the testing phase, GANomaly detects anomalies by scoring test data. The anomaly score is calculated by subtracting the anomaly score of the normal data from that of the abnormal data. Generally, abnormal data have a much higher anomaly score than normal data, making it possible to identify anomalies based on this score. The anomaly score is calculated for each sample using the encoder loss [46]. Therefore, during training, the anomaly scores were analyzed by comparing experiments, placing more emphasis on the weight of the encoder loss (W_{enc}) than on the other weight loss functions. The anomaly score $A(x)$ for x is defined as follows:

$$A(x) = \|G_E(x) - E(G(x))\|_1 \quad (5)$$

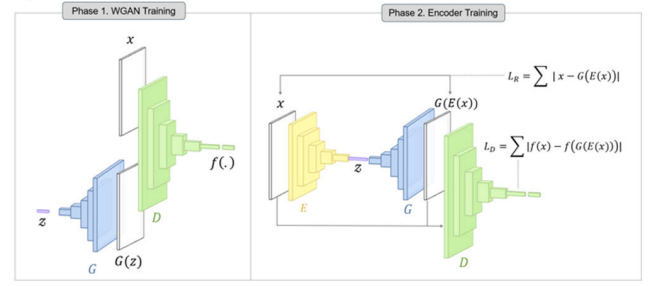


FIGURE 21. f-AnoGAN model framework.

2) F-ANOGAN

The f-AnoGAN [15] model is based on GAN and is known to be one of the most effective models for unsupervised anomaly detection, similar to GANomaly. This model is an improved version of AnoGAN, which presents the challenge of slow speed in the iterative mapping process of converting input image data into latent space. However, f-AnoGAN addresses this issue by utilizing an encoder to map images directly into latent space, resulting in faster performance. f-AnoGAN consists of three subnetworks and is trained in two stages, as shown in Fig. 21.

In the initial training phase, the generator G and discriminator D are trained using normal data to learn the latent space. Throughout the training process, the generator G takes a noise input vector z sampled from the latent space Z and learns to map it into image space X , whereas discriminator D assesses the similarity of the distributions of the input data x and generated data $G(z)$ by outputting a value. The generator G and discriminator D that are trained in this step are also used in the next step.

In the second training phase, encoder E is trained to map the input data x to the learned latent space z , whereas generator G is trained to map it back to the image space. During this process, the discriminator D is responsible for distinguishing between the original input data x and reconstructed data $G(E(x))$.

The loss function L_{izif} of f-AnoGAN is as follows:

$$L_{izif} = \frac{1}{n} \|x - G(E(x))\|_2 + \frac{k}{n_d} \|f(x) - f(G(E(x)))\|_2 \quad (6)$$

where n_d is the dimension of the intermediate feature representation, and k is the weighting factor:

Anomaly detection in the f-AnoGAN follows a process similar to that of encoder training. First, encoder E maps the test data to latent space, and generator G maps the latent representation to image space. Anomalies are identified by calculating an anomaly score that measures the difference between the reconstructed data $G(E(x))$ and the original input x . In contrast to GANomaly, f-AnoGAN employs a single loss function, allowing experiments to be conducted to observe changes in the anomaly score by adjusting the weighting factor of this loss function. The anomaly score is defined as follows:

$$A(x) = A_R(x) + k \cdot A_D(x)_1 \quad (7)$$

where $A_R(x) = \frac{1}{n} \|x - G(E(x))\|_2$, $A_D(x) = \frac{1}{n_d} \|f(x) - f(G(E(x)))\|_2$, and k is the weighting factor.

IV. EXPERIMENTS

As mentioned in Section II-C, we conducted experiments using the GANomaly and f-AnoGAN models to detect deep-fake audio through unsupervised learning. We analyzed the variations in the Mel spectrograms and MFCCs based on the number of Mel filters.

Both models were trained on real audio data, with the objective of distinguishing between untrained real and deep-fake audio. This approach enables the immediate detection of a wide range of deep fake audios that continue to be generated.

The hyperparameter tuning for each model in the experiments are shown in the table 5. Hyperparameter tuning refers to the process of identifying and selecting the optimal hyperparameters for training a machine learning model. When performed effectively, hyperparameter tuning minimizes the loss function, thereby maximizing performance and ensuring the highest possible accuracy.

TABLE 5. Parameter settings for each model.

Hyperparameter	GANomaly	f-AnoGAN
batch size	64	32
optimizer	adam	adam
epoch	[1, 800]	[1, 500]
loss function	adv, cnt, enc	Lizif
loss function Weight	[1, 50]	[1, 50]

The initial setup for the GANomaly experiment involved setting the number of Mel filters for the mel-spectrogram and MFCC features, with the filter count set to 20, 40, and 128. The models were trained for the initial 100 epochs, in increments of 100 epochs. After selecting the model that demonstrated the highest performance for each preprocessing configuration, we adjusted the weights of the GANomaly loss functions (contextual loss, adversarial loss, and encoder loss) based on the optimal number of training epochs and evaluated the performance changes according to these weights.

The f-AnoGAN experiments were initially configured under the same conditions as those of the GANomaly experiments. However, as mentioned in Sections III D, considering the distinct loss function utilized by f-AnoGAN, we adjusted the kappa coefficient to examine its impact on the performance.

In Section III-A, the highest performance achieved for each case is documented in table 6 and 7 with the metrics including Accuracy, F1 score, PR-Curve, PR AUC, ROC-Curve, and ROC-AUC. The analysis focused on the F1 score, ROC curve, and partial response (PR) curve. Section III-B provides a detailed analysis of the results under various exper-

imental conditions and offers interpretations from a societal perspective.

A. RESULT

The experimental results for GANomaly and f-AnoGAN are shown below.

1) GANOMALY

The results of the experiments conducted with GANomaly using the mel-spectrogram and MFCC with mel filter counts of 20, 40, and 128 are as follows: In GANomaly, the mel spectrogram with a mel filter count of 40 achieved the highest F1 score of 0.93, and this result was obtained without applying any weighting.

2) F-ANOGAN

The experiments conducted with f-AnoGAN using the mel-spectrogram, and MFCC with mel filter counts of 20, 40, and 128 yielded the following results, With f-AnoGAN, the MFCC with a mel filter count of 20 achieved the highest F1 score of 0.88, which was obtained with a kappa value of 10.

The F1 score was adopted as the primary metric to represent the overall performance in our experiments. The F1 score is the harmonic mean of Precision and Recall, effectively evaluating the balance between these two metrics. By considering both false positives and false negatives, the F1 score provides a more accurate assessment of the model performance, particularly on imbalanced datasets, making it an appropriate metric for our study. Precision refers to the proportion of correctly identified deepfake (abnormal) audio among all instances predicted as deepfake by the model, whereas Recall refers to the proportion of actual deepfake audio that the model correctly predicts as deepfake.

Using the Mel-spectrogram with 40 Mel filters, we achieved an F1 score of 0.93, indicating that both Precision and Recall were above a certain threshold. This suggests that the model effectively and comprehensively analyzes the deepfake audio.

To evaluate the overall performance of the binary classification system across all thresholds, we used not only the F1 score but also the EER, along with the ROC and PR curves. EER represents the point in a binary classification system where the False Acceptance Rate (FAR) equals the False Rejection Rate (FRR). It evaluates the model's error balance across different thresholds, thereby reflecting the system's balanced performance. Unlike the F1 score, which reflects performance at a specific threshold, ROC and PR curves provide insight into the model's stability by illustrating performance across the entire all threshold spectrum.

The PR curve for GANomaly shown in Fig. 22 reveals that the highest performance reached 0.98, which is very high. This indicates that the model effectively identified deepfake audios with a minimum number of false positives (FP) and false negatives (FN).

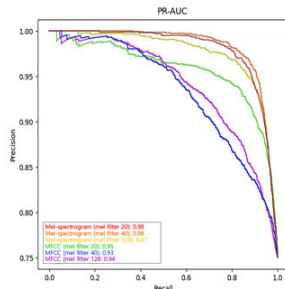
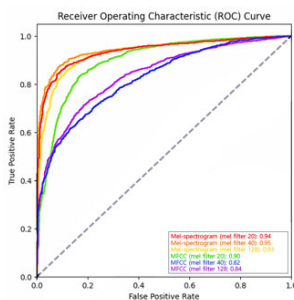
Similarly, the ROC curve for GANomaly shown in Fig. 23 indicates that the highest performance reached 0.95, which

TABLE 6. Evaluation metrics of the GANomaly model for each case.

Pre-processing	Mel filters	Epochs	Weighted loss	ROC	ACC	F1	PR	EER
Mel-spectrogram	20	200	no weight	0.94	0.88	0.92	0.98	0.12
Mel-spectrogram	40	300	no weight	0.95	0.89	0.93	0.98	0.10
Mel-spectrogram	128	200	no weight	0.93	0.88	0.92	0.97	0.13
MFCC	20	500	adv=50	0.9	0.86	0.90	0.95	0.16
MFCC	40	700	no weight	0.82	0.78	0.87	0.93	0.26
MFCC	128	300	cnt=50	0.84	0.8	0.87	0.94	0.24

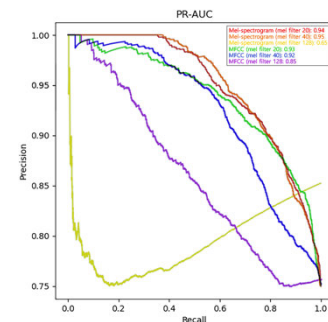
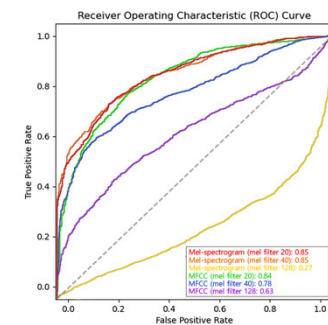
TABLE 7. Evaluation metrics of the f-AnoGAN model for each case.

Pre-processing	Mel filters	Epochs	Weighted loss	ROC	ACC	F1	PR	EER
Mel-spectrogram	20	200	kappa=10	0.85	0.80	0.87	0.94	0.16
Mel-spectrogram	40	100	kappa=50	0.85	0.80	0.87	0.94	0.14
Mel-spectrogram	128	200	kappa=1	0.71	0.49	0.66	0.57	0.27
MFCC	20	100	kappa=10	0.84	0.81	0.88	0.93	0.15
MFCC	40	200	kappa=50	0.76	0.77	0.86	0.90	0.20
MFCC	128	100	kappa=10	0.63	0.66	0.79	0.85	0.25

**FIGURE 22.** PR-Curve of the optimal GANomaly model for each case.**FIGURE 23.** ROC-Curve of the optimal GANomaly model for each case.

is also a very high level. This demonstrates that the model achieved both a high true positive rate (TPR) and a low false positive rate (FPR), indicating its robustness against decision threshold variations.

As shown in Figs. 24 and 25, the GANomaly model using a mel-spectrogram with 40 mel filters and without weight adjustment, achieved the highest performance in terms of F1 score, ROC, and PR curves with values of 0.93, 0.95, and 0.98, respectively. This indicates that the model reliably detects both deepfake and real audios at specific thresholds as well as across thresholds.

**FIGURE 24.** PR-Curve of the optimal f-AnoGAN model for each case.**FIGURE 25.** ROC-Curve of the optimal f-AnoGAN model for each case.

B. ANALYZING EXPERIMENTS

In this section, we analyze performance based on the number of mel filters, training iterations, and weight allocations. Subsequently, the configurations that achieve the highest performances are described.

1) ANALYZING PERFORMANCE BY NUMBER OF MEL FILTERS AND NUMBER OF TRAININGS

Based on the analysis of the performance with respect to the number of Mel filters, Fig. 26 indicates that there are 20 filters

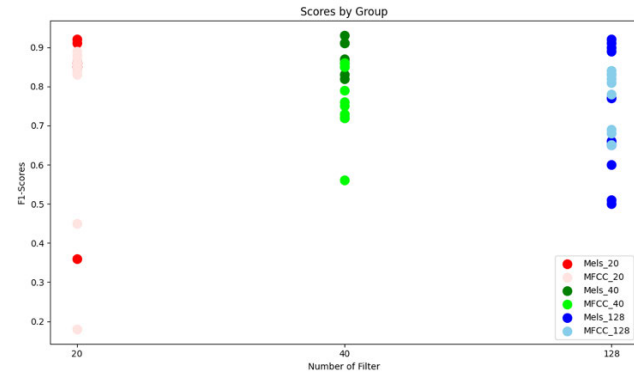


FIGURE 26. Score distribution by number of mel filters.

TABLE 8. Score table based on number of training repetitions.

Epochs	Average	Median	High F1 score
100	0.7525	0.835	0.91
200	0.8	0.855	0.92
300	0.766	0.835	0.93
400	0.793	0.835	0.9
500	0.784	0.835	0.89

in the red color range, 40 filters in the green color range, and 128 filters in the blue color range.

The analysis showed that the highest performance was achieved when using 40 Mel filters, indicating the importance of the number of Mel filters in audio feature extraction. The superior results with 40 Mel filters, as opposed to 20 or 128, can be attributed to the optimal frequency band division, which captures more details of the audio signal. As mentioned in Section II-C, previous research on speech recognition typically employed 20–40 mel filters, and our findings further support that 40 mel filters strike an ideal balance for deepfake audio detection tasks. Additionally, mel-spectrogram preprocessing, which maps audio signals to the primary colors (red, blue, and green), demonstrated higher accuracy compared to MFCC. This highlights the advantage of Mel-Spectrograms over MFCC in deep-fake detection, suggesting that mel-spectrogram-based approaches are more effective for speech-signal processing. The results provide a promising basis for future advancements in speech recognition and deep-fake audio.

A performance analysis based on the number of training iterations revealed that at 200 iterations, the average F1 score peaked at 0.8, with the highest median value recorded at 0.855. As shown in table 8 and Fig. 27, this indicates that at 200 training iterations, the model predictions were relatively consistent, with an evenly distributed performance.

In addition, after 300 training iterations, the model achieved its highest F1 score. This suggests that the optimal range for maximizing the model performance lies between 200 and 300 iterations, which underscores the critical role of selecting an appropriate number of training iterations in enhancing model performance and offers valuable insights into the relationship between iteration count and model performance.

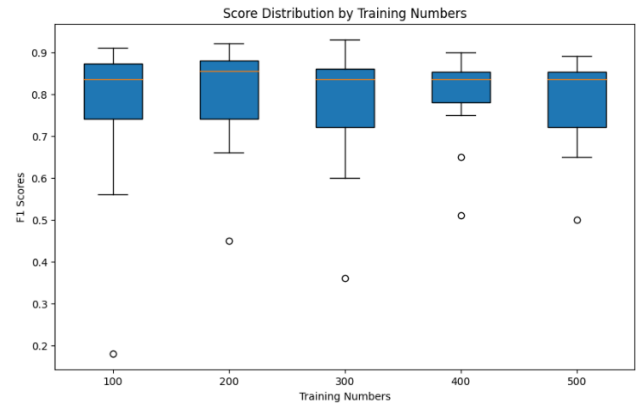


FIGURE 27. Boxplot based on number of training iterations.

2) ANALYZE PERFORMANCE AS LOSS FUNCTION WEIGHT COEFFICIENT CHANGE

The weight coefficient of the loss function experiment was analyzed based on the optimal number of training iterations determined in the previous experiment.

For GANomaly, as mentioned in Section III D, the anomaly score was calculated for all samples using encoder loss. To explore the impact of varying the loss-coefficient weights, including the encoder loss weight W_{enc} in (5), a series of experiments were performed. Fig. 28 presents a box plot of the model performance across different weight coefficients, with the red line representing the average performance for each configuration. When the adversarial loss weight coefficient is prioritized, the model achieves an average performance of 0.7. Shifting the focus to the contextual loss weight coefficient resulted in a slightly higher average of 0.77, whereas emphasizing the encoder loss weight yielded an average performance of 0.795. However, the highest performance of 0.895 was observed when no specific weight coefficient was applied, indicating that the model performed best when the loss function weights were balanced naturally without assigning any weight coefficient.

The experiments revealed that when a weighting coefficient is applied, the encoder loss function yields strong performance, and the best results were achieved when no explicit weighting was applied to the loss functions. Although the anomaly score in GANomaly is based on the encoder loss formula, adjusting the encoder loss weight coefficient during training does not lead to better performance.

In the case of the f-AnoGAN, as mentioned in Section III-D2, a single loss function is used, and the performance of the model is evaluated based on changes in the weight coefficient (κ). Fig. 29 shows a box plot comparing the F1 scores when the weight coefficient (κ) value is set to the default value of 1 and when it is increased (turned). The red line represents the average F1 score for each configuration. The average F1 score is 0.82, with a default coefficient(κ) value of 0.75 when the weight coefficient(κ) is increased, indicating that the highest performance is achieved with the default setting. However,

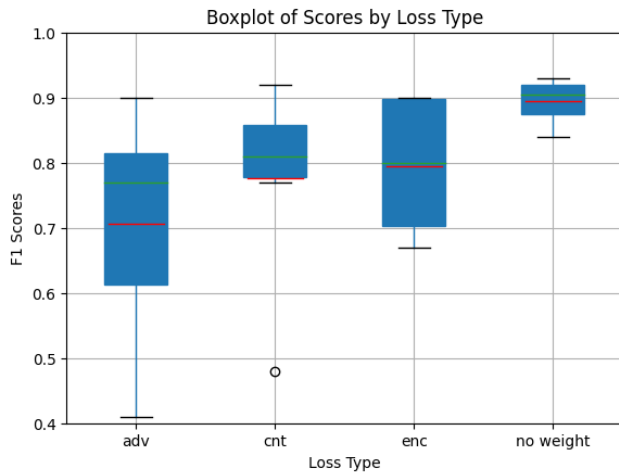


FIGURE 28. Boxplot based on GANomaly loss function weight change.

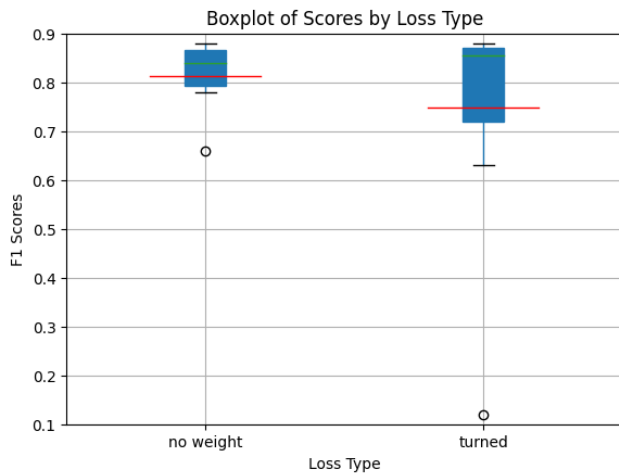


FIGURE 29. Boxplot based on f-AnoGAN loss function weight change.



FIGURE 30. Distribution of anomaly scores in training data.

as discussed in Section IV-A2, the optimal weight coefficient (κ) value in f-AnoGAN does not have a significant impact on performance.

3) ANOMALY SCORE DISTRIBUTION

To evaluate the performance of the anomaly detection model, we analyzed the distribution of the anomaly scores for both the training and test datasets. The anomaly score quantifies the difference in the data distribution, with values closer to 0 indicating a higher similarity to the training data of genuine audio samples.

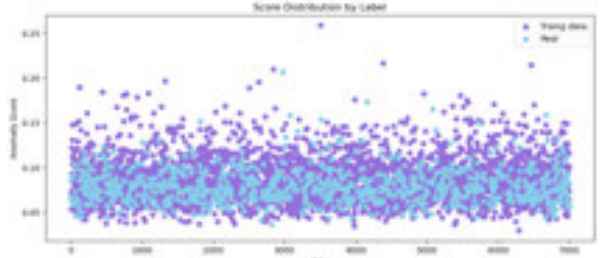


FIGURE 31. Distribution of anomaly scores in training data and untrained normal data.

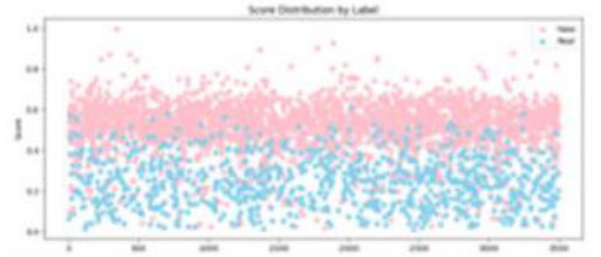


FIGURE 32. Distribution of anomaly scores in untrained normal and untrained anomalous data.



FIGURE 33. Distribution when trained on GANomaly using mel spectrogram with mel filter value of 40.

Fig. 30 presents the distribution of the anomaly scores for training the real audio data, where the scores are clustered near 0. This indicates that the model effectively learned the patterns of the training data, which were genuine audio samples.

Fig. 31 presents a comparison of the anomaly score distributions of the training real audio data and untrained real audio data. The similarity between these distributions demonstrates that the model identified the untrained real audio data as genuine, reflecting its ability to generalize to previously unseen real audio samples. This strong generalization capability is a key indicator of the robustness of the model in recognizing authentic audio.

Fig. 32 presents the anomaly score distributions for the untrained real audio data and untrained anomalous deepfake audio data. The distributions indicate that the anomaly scores for the untrained deepfake audio are significantly higher than those for the untrained real audio. This demonstrates the ability of the model to effectively distinguish between genuine and anomalous deep-fake audio, even when data were not seen during training.

The scatter plots of the anomaly scores for untrained real audio test data and untrained deepfake audio data in

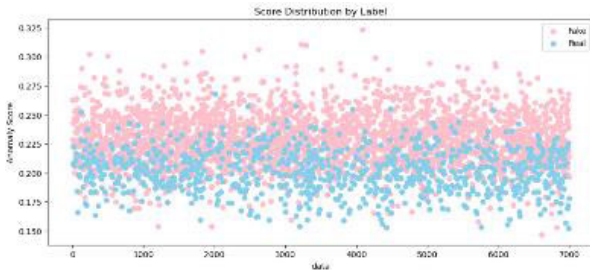


FIGURE 34. Distribution when trained on f-AnoGAN using MFCC with mel filter value of 20.

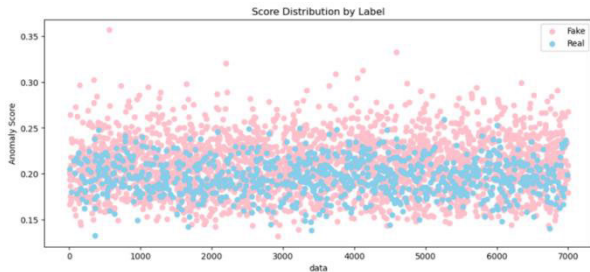


FIGURE 35. Distribution for f-AnoGAN trained using MFCC with mel filter value 128.



FIGURE 36. Distribution when trained on GANomaly using mel spectrogram with mel filter value of 40.



FIGURE 37. Distribution when trained on GANomaly using mel spectrogram with mel filter value of 128.

GANomaly and f-AnoGAN are shown in Figs. 33 and 34, respectively. In GANomaly, using 40 Mel filters with a Mel spectrogram (Fig. 33), and in f-AnoGAN, using 20 Mel filters with MFCC (Fig. 34), both plots demonstrate minimal overlap between the distribution of points for untrained deepfake audio and untrained real audio. However, GANomaly shows superior discrimination performance compared with f-AnoGAN, indicating that GANomaly captures and distinguishes subtle differences between genuine and deepfake audio samples.

Furthermore, in f-AnoGAN, when using MFCC with 128 Mel filters (Fig. 35), the scatterplot reveals a significant

TABLE 9. F1 score and recall for mel filters in GANomaly.

	mel filter = 40	mel filter = 128
F1 Score	0.930	0.923
Recall	0.925	0.940

overlap between the distributions of deepfake and real audio. This suggests that the model struggles to effectively capture the distinguishing characteristics of the audio, particularly the differences between deepfake and real audio, thereby failing to extract sufficient discriminative information from the audio signal, leading to a reduced performance in identifying deepfake audio.

Fig. 36 presents the scatterplot for GANomaly trained using a mel spectrogram with 40 features, whereas Fig. 37 shows the scatterplot for GANomaly trained with 128 features. As shown in Table 9, the F1 score for Fig. 36 is 0.93, which is slightly higher than the F1 score of 0.92 corresponding to Fig. 37. However, the recall value is higher for Fig. 37. Upon closer inspection of the scatterplots, we observe that the distribution of deepfake audio overlaps more with the real audio in Fig. 36 than in Fig. 37. This indicates that a deepfake audio is less likely to be misclassified as real when 128 features are used, which is further supported by the higher recall value in Fig. 37.

Because our research focuses on selecting the optimal model for preventing crimes related to deepfake audio-based fraud and impersonation, consider both the social and technical aspects is essential. From a social perspective, the optimal model should prioritize minimizing false negatives, ensuring that deep-fake audio recordings are not mistakenly classified as real. As demonstrated by the recall values and scatter plots, this evaluation is crucial for mitigating the risks associated with misuse of deep fake audio. However, we must also account for the possibility of false positives in which real speech is mistakenly identified as deepfake audio. Therefore, quantitative aspects, including metrics, such as F1 scores and precision, should be considered to ensure a balanced and reliable detection system.

4) COMPARISON OF QUANTITATIVE RESULTS FOLLOWING OPTIMAL HYPERPARAMETER-TUNING

The optimal results were achieved by hyperparameter tuning of the number of mel filters, number of training iterations, and weighting of the loss function. The refined models are illustrated in Fig. 38 for GANomaly and Fig. 39 for f-AnoGAN, demonstrating their respective performance improvements after these adjustments.

For GANomaly, the overall performance using the mel spectrogram is distributed in the range of 0.85-0.9, demonstrating a higher performance compared to the distribution observed with MFCC. In addition, similar to the results obtained before hyperparameter tuning, the best performance was achieved when the number of Mel filters was set to 40. This indicates that the optimal configuration for this model is attained with 40 mel filters.

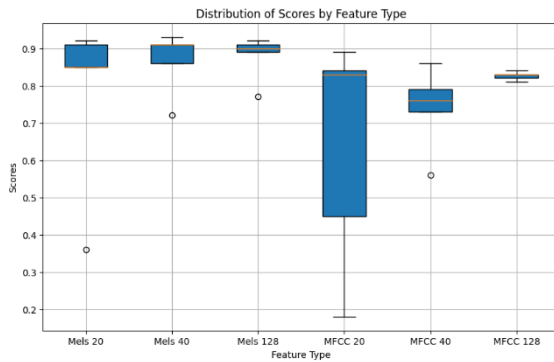


FIGURE 38. Boxplot of GANomaly F1 Score for each case.

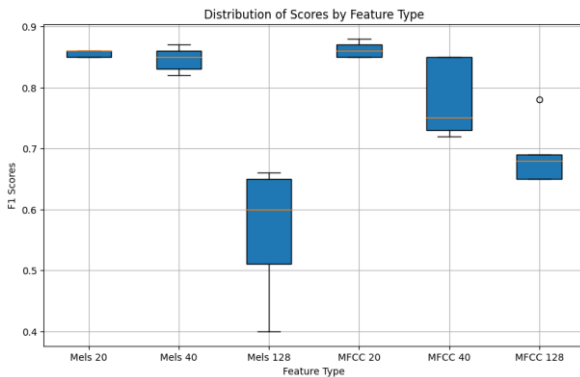


FIGURE 39. Boxplot of f-AnoGAN F1 Score for each case.

In the case of the f-AnoGAN shown in Fig. 39, the performance of the model was influenced by the number of mel filters. Notably, the performance declined when the number of mel filters was set to 128 and better results were observed when fewer filters were used. The highest performance was achieved with 20 Mel filters, indicating that a lower number of mel filters is more effective for this model.

Although the optimal number of mel filters may vary depending on the preprocessing method, our results indicate that a smaller number of filters generally leads to higher performance. In this study, the optimal number of mel filters was determined to be 40, suggesting that setting the number of mel filters to 40 can significantly improve the performance when designing a synthetic speech detection system.

Additionally, our experiments demonstrated that the GANomaly model using the mel spectrogram performs effectively in deep-fake audio detection, achieving an F1 score of 0.93.

Furthermore, because ASVspoof 2021 trained on real speech data, it successfully detected ASVspoof 2019's real speech data and WaveFake's synthetic speech data generated using different techniques, validating the usefulness of unsupervised methods in identifying new types of synthetic speech. This highlights the potential of these methods for deepfake audio detection.

V. CONCLUSION

We conducted comprehensive experiments using two speech preprocessing methods and two unsupervised learning mod-

els, resulting in four distinct combinations. For speech preprocessing, we varied the number of mel filters in both the mel-spectrogram and MFCC to 20, 40, and 128, and identified the optimal epochs for each filter configuration to effectively capture human speech features. Subsequently, we adjusted the loss-function weights in the GANomaly and f-AnoGAN models based on the best-performing epochs to determine the most effective parameter settings.

In conclusion, the mel-spectrogram preprocessing method with 40 mel filters and 300 epochs, combined with GANomaly without loss function weight adjustment, yielded the highest performance, with an F1 score of 0.93. This configuration proved to be the most effective for deepfake audio detection.

However, mistaking deep audio for real audio can have serious consequences. To address this issue, it is crucial to consider the recall value in the evaluation. The combination of the mel-spectrogram with 40 mel filters and 300 epochs, which was identified as the best-performing configuration, achieved a recall value of 0.925.

On the other hand, using 128 mel filters with 200 epochs and leaving the GANomaly loss function weight unchanged yielded an F1 score of 0.923, which is slightly lower than the best result. However, the recall value is higher at 0.940. This finding suggests that further discussion is required to balance quantitative performance metrics with broader considerations, such as minimizing false negatives in socially critical applications.

This study highlights the potential of using an unsupervised anomaly detection approach for identifying deepfake audio. By learning the characteristics of genuine human speech, the model can classify other voices as deepfakes without additional training, making it particularly effective for detecting new types of synthetic audio. This method has a range of applications. For instance, it can enhance the detection and prevention of voice phishing scams by quickly identifying deepfake audio in voice-enabled communication systems such as voice calls or video chats. In identity verification processes, such as account opening or card issuance, this method can help block fraudulent transactions by third parties using manipulated audio. Furthermore, it can quickly determine if a voice of a celebrity or influential individual has been altered, enabling rapid responses to misinformation or fake news. The technique is also valuable for detecting voice manipulation in scenarios like preventing voice theft and mitigating associated damages. We anticipate that this research will contribute to significant advancements in the field of unsupervised deepfake audio detection.

ACKNOWLEDGMENT

(Daeun Song, Nayoung Lee, and Jiwon Kim are co-first authors.)

REFERENCES

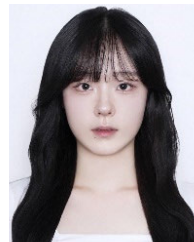
- [1] A. Miotti and A. Wasil, "Combating deepfakes: Policies to address national security threats and rights violations," 2024, *arXiv:2402.09581*.

- [2] M. Oancea, "Ai and deep fake-video and audio manipulation techniques capable of altering the political process," *Revista de Stiinta Politice. Revue des Sci. Politiques*, vol. 81, no. 5, pp. 70–82, 2024.
- [3] C. E. Steck and A. Kaczynski. (2024). *Fake Joe Biden Robocall Urges New Hampshire Voters Not To Vote in Tuesday's Democratic Primary*. [Online]. Available: <https://edition.cnn.com/2024/01/22/politics/fake-joe-bidenrobocall/index.html>
- [4] J. Twomey, D. Ching, M. P. Aylett, M. Quayle, C. Linehan, and G. Murphy, "Do deepfake videos undermine our epistemic trust? A thematic analysis of tweets that discuss deepfakes in the Russian invasion of Ukraine," *PLoS ONE*, vol. 18, no. 10, Oct. 2023, Art. no. e0291668, doi: [10.1371/journal.pone.0291668](https://doi.org/10.1371/journal.pone.0291668).
- [5] A. Hamza, A. R. R. Javed, F. Iqbal, N. Kryvinska, A. S. Almadhor, Z. Jalil, and R. Borghol, "Deepfake audio detection via MFCC features using machine learning," *IEEE Access*, vol. 10, pp. 134018–134028, 2022, doi: [10.1109/ACCESS.2022.3231480](https://doi.org/10.1109/ACCESS.2022.3231480).
- [6] I. Altalihin, S. AlZu'bi, A. Alqudah, and A. Mughaid, "Unmasking the truth: A deep learning approach to detecting deepfake audio through MFCC features," in *Proc. Int. Conf. Inf. Technol. (ICIT)*, Aug. 2023, pp. 511–518.
- [7] R. L. M. A. P. C. Wijethunga, D. M. K. Matheesha, A. A. Noman, K. H. V. T. A. De Silva, M. Tissera, and L. Rupasinghe, "Deepfake audio detection: A deep learning based solution for group conversations," in *Proc. 2nd Int. Conf. Advancements Comput. (ICAC)*, vol. 1, Dec. 2020, pp. 192–197, doi: [10.1109/ICAC51239.2020.9357161](https://doi.org/10.1109/ICAC51239.2020.9357161).
- [8] T. M. Wani, S. A. A. Qadri, D. Communiello, and I. Amerini, "Detecting audio deepfakes: Integrating CNN and BiLSTM with multi-feature concatenation," in *Proc. ACM Workshop Inf. Hiding Multimedia Secur.*, Jun. 2024, pp. 271–276, doi: [10.1145/3658664.3659647](https://doi.org/10.1145/3658664.3659647).
- [9] B. Kumar and S. R. Alraisi, "Deepfakes audio detection techniques using deep convolutional neural network," in *Proc. Int. Conf. Mach. Learn., Big Data, Cloud Parallel Comput. (COM-IT-CON)*, vol. 1, May 2022, pp. 463–468, doi: [10.1109/COM-IT-CON54601.2022.9850771](https://doi.org/10.1109/COM-IT-CON54601.2022.9850771).
- [10] P. Laskov, "Learning intrusion detection: Supervised or unsupervised," in *Proc. 13th Int. Conf.*, 2005, pp. 50–57.
- [11] Y. Patel, S. Tanwar, R. Gupta, P. Bhattacharya, I. E. Davidson, R. Nyameko, S. Aluvala, and V. Vimal, "Deepfake generation and detection: Case study and challenges," *IEEE Access*, vol. 11, pp. 143296–143323, 2023, doi: [10.1109/ACCESS.2023.3342107](https://doi.org/10.1109/ACCESS.2023.3342107).
- [12] V. Chandola, A. Banerjee, and V. Kumar, "Anomaly detection: A survey," *ACM Comput. Surveys*, vol. 41, no. 3, pp. 1–58, Jul. 2009.
- [13] A. Jiang, W.-Q. Zhang, Y. Deng, P. Fan, and J. Liu, "Unsupervised anomaly detection and localization of machine audio: A gan-based approach," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Jun. 2023, pp. 1–5, doi: [10.1109/ICASSP49357.2023.10096813](https://doi.org/10.1109/ICASSP49357.2023.10096813).
- [14] J. Shim, T. Joung, S. Lee, and J.-I. Ha, "Audio data-driven anomaly detection for induction motor based on generative adversarial networks," in *Proc. IEEE Energy Convers. Congr. Expo. (ECCE)*, Oct. 2022, pp. 1–5, doi: [10.1109/ECCE50734.2022.9947652](https://doi.org/10.1109/ECCE50734.2022.9947652).
- [15] T. Schlegl, P. Seeböck, S. M. Waldstein, G. Langs, and U. Schmidt-Erfurth, "F-AnoGAN: Fast unsupervised anomaly detection with generative adversarial networks," *Med. Image Anal.*, vol. 54, pp. 30–44, May 2019, doi: [10.1016/j.media.2019.01.010](https://doi.org/10.1016/j.media.2019.01.010).
- [16] E. A. AlBadawy, "Detecting AI-synthesized speech using bispectral analysis," in *Proc. CVPR Workshops*, 2019, pp. 104–109.
- [17] A. Ustubioglu, B. Ustubioglu, and G. Ulutas, "Mel spectrogram-based audio forgery detection using CNN," *Signal, Image Video Process.*, vol. 17, no. 5, pp. 2211–2219, Jul. 2023, doi: [10.1007/s11760-022-02436-4](https://doi.org/10.1007/s11760-022-02436-4).
- [18] S. Seo, C. Kim, and J.-H. Kim, "Convolutional neural networks using log mel-spectrogram separation for audio event classification with unknown devices," *J. Web Eng.*, vol. 21, pp. 497–522, Jan. 2022.
- [19] F. Zheng, G. Zhang, and Z. Song, "Comparison of different implementations of MFCC," *J. Comput. Sci. Technol.*, vol. 16, no. 6, pp. 582–589, Nov. 2001, doi: [10.1007/bf02943243](https://doi.org/10.1007/bf02943243).
- [20] A. Pianese, D. Cozzolino, G. Poggi, and L. Verdoliva, "Deepfake audio detection by speaker verification," in *Proc. IEEE Int. Workshop Inf. Forensics Security*, Dec. 2022, pp. 1–6, doi: [10.1109/WIFS55849.2022.9975428](https://doi.org/10.1109/WIFS55849.2022.9975428).
- [21] N. M. Muller, F. Dieckmann, and J. Williams, "Attacker attribution of audio deepfakes," 2022, *arXiv:2203.15563*.
- [22] Z. Kang, Y. He, B. Zhao, X. Qu, J. Peng, J. Xiao, and J. Wang, "Retrieval-augmented audio deepfake detection," in *Proc. Int. Conf. Multimedia Retr.*, May 2024, pp. 376–384, doi: [10.1145/3652583.3658086](https://doi.org/10.1145/3652583.3658086).
- [23] Y. Guo, H. Huang, X. Chen, H. Zhao, and Y. Wang, "Audio deepfake detection with self-supervised wavlm and multi-fusion attentive classifier," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Apr. 2024, pp. 12702–12706.
- [24] P. Kawa, M. Plata, and P. Syga, "Attack agnostic dataset: Towards generalization and stabilization of audio deepfake detection," 2022, *arXiv:2206.13979*.
- [25] A. Firc, K. Malinka, and P. Hanáček, "Deepfake speech detection: A spectrogram analysis," in *Proc. 39th ACM/SIGAPP Symp. Appl. Comput.*, Apr. 2024, pp. 1312–1320.
- [26] J. Frank and L. Schönherr, "WaveFake: A data set to facilitate audio deepfake detection," 2021, *arXiv:2111.02813*.
- [27] N. M. Müller, P. Kawa, S. Hu, M. Neu, J. Williams, P. Sperl, and K. Böttinger, "A new approach to voice authenticity," 2024, *arXiv:2402.06304*.
- [28] Z. Cai and M. Li, "Integrating frame-level boundary detection and deepfake detection for locating manipulated regions in partially spoofed audio forgery attacks," *Comput. Speech Lang.*, vol. 85, Apr. 2024, Art. no. 101597, doi: [10.1016/j.csl.2023.101597](https://doi.org/10.1016/j.csl.2023.101597).
- [29] M. Todisco, X. Wang, V. Vestman, M. Sahidullah, H. Delgado, A. Nautsch, J. Yamagishi, N. Evans, T. Kinnunen, and K. Aik Lee, "ASVspoof 2019: Future horizons in spoofed and fake audio detection," 2019, *arXiv:1904.05441*.
- [30] X. Wang et al., "ASVspoof 2019: A large-scale public database of synthesized, converted and replayed speech," *Comput. Speech Lang.*, vol. 64, Nov. 2020, Art. no. 101114, doi: [10.1016/j.csl.2020.101114](https://doi.org/10.1016/j.csl.2020.101114).
- [31] J. Yamagishi, X. Wang, M. Todisco, M. Sahidullah, J. Patino, A. Nautsch, X. Liu, K. A. Lee, T. Kinnunen, N. Evans, and H. Delgado, "ASVspoof 2021: Accelerating progress in spoofed and deepfake speech detection," in *Proc. Ed. Autom. Speaker Verification Spoofing Countermeasures Challenge*, Sep. 2021, pp. 1–20.
- [32] X. Liu, X. Wang, M. Sahidullah, J. Patino, H. Delgado, T. Kinnunen, M. Todisco, J. Yamagishi, N. Evans, A. Nautsch, and K. A. Lee, "ASVspoof 2021: Towards spoofed and deepfake speech detection in the wild," *IEEE/ACM Trans. Audio, Speech, Language Process.*, vol. 31, no. 2, pp. 2507–2522, Apr. 2021, doi: [10.1109/TASLP.2023.3285283](https://doi.org/10.1109/TASLP.2023.3285283).
- [33] R. Reimao and V. Tzerpos, "FoR: A dataset for synthetic speech detection," in *Proc. Int. Conf. Speech Technol. Hum.-Comput. Dialogue (SpED)*, Oct. 2019, pp. 1–10.
- [34] I. Solak. (2019). *The M-AILABS Speech Dataset*. [Online]. Available: <https://github.com/imdatcelest/m-ailabs-dataset>
- [35] J. Yi, R. Fu, J. Tao, S. Nie, H. Ma, C. Wang, T. Wang, Z. Tian, Y. Bai, C. Fan, S. Liang, S. Wang, S. Zhang, X. Yan, L. Xu, Z. Wen, and H. Li, "ADD 2022: The first audio deep synthesis detection challenge," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2022, pp. 9216–9220, doi: [10.1109/ICASSP43922.2022.9746939](https://doi.org/10.1109/ICASSP43922.2022.9746939).
- [36] R. Yan, C. Wen, S. Zhou, T. Guo, W. Zou, and X. Li, "Audio deepfake detection system with neural stitching for ADD 2022," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2022, pp. 9226–9230, doi: [10.1109/ICASSP43922.2022.9746820](https://doi.org/10.1109/ICASSP43922.2022.9746820).
- [37] A. K. Singh and P. Singh, "Detection of AI-synthesized speech using cepstral & bispectral statistics," in *Proc. IEEE 4th Int. Conf. Multimedia Inf. Process. Retr. (MIPR)*, Sep. 2021, pp. 412–417, doi: [10.1109/MIPR51284.2021.00076](https://doi.org/10.1109/MIPR51284.2021.00076).
- [38] A. Fathan, J. Alam, and W. H. Kang, "Mel-spectrogram image-based end-to-end audio deepfake detection under channel-mismatched conditions," in *Proc. IEEE Int. Conf. Multimedia Expo. (ICME)*, Jul. 2022, pp. 1–6, doi: [10.1109/ICME52920.2022.9859621](https://doi.org/10.1109/ICME52920.2022.9859621).
- [39] M. Mcuba, A. Singh, R. A. Ikuesan, and H. Venter, "The effect of deep learning methods on deepfake audio detection for digital investigation," *Proc. Comput. Sci.*, vol. 219, pp. 211–219, May 2023, doi: [10.1016/j.procs.2023.01.283](https://doi.org/10.1016/j.procs.2023.01.283).
- [40] N. Mahmoodian, A. Schaufler, A. Pashazadeh, A. Boese, M. Friebe, and A. Illanes, "Proximal detection of guide wire perforation using feature extraction from bispectral audio signal analysis combined with machine learning," *Comput. Biol. Med.*, vol. 107, pp. 10–17, Apr. 2019, doi: [10.1016/j.combiomed.2019.02.001](https://doi.org/10.1016/j.combiomed.2019.02.001).
- [41] S. S. Abeysekera, "Multiple pitch estimation of poly-phonic audio signals in a frequency-lag domain using the bispectrum," in *Proc. IEEE Int. Symp. Circuits Syst.*, Jun. 2004, p. 469, doi: [10.1109/ISCAS.2004.1328785](https://doi.org/10.1109/ISCAS.2004.1328785).
- [42] C. L. Nikias and M. R. Raghuveer, "Bispectrum estimation: A digital signal processing framework," *Proc. IEEE*, vol. 75, no. 7, pp. 869–891, 1987.
- [43] E. A. A. Abdrabuh. (2022). *AI-Synthesized Speech: Generation and Detection*. [Online]. Available: <https://www.proquest.com/dissertations-theses/ai-synthesized-speech-generation-detection/docview/2661971298/se-2>
- [44] B. Zhang. (2019). *Audio Recognition Using Mel Spectrograms and Convolution Neural Networks*. [Online]. Available: <https://api.semanticscholar.org/CorpusID>

- [45] C. Ittichaichareon, "Speech recognition using MFCC," in *Proc. Int. Conf. Comput. Graph., Simul. Model.*, vol. 9, 2012, pp. 1–20.
- [46] S. Akcay, A. Atapour-Abarghouei, and T. P. Breckon, "GANomaly: Semi-supervised anomaly detection via adversarial training," in *Proc. Asian Conf. Comput. Vis.*, Dec. 2018, pp. 622–637.
- [47] W. Zhu, "Unsupervised anomalous sound detection via semi-supervised GANomaly adversarial training," Detection Classification Acoust. Scenes Events Challenge, 2021. [Online]. Available: https://dcase.community/documents/challenge2021/technical_reports/DCASE2021_Liu_13_t2.pdf
- [48] W. Yang and C. Zeng, "A hybrid anomaly detection model based on GANomaly in cloud environment," in *Proc. IEEE 5th Int. Conf. Big Data Artif. Intell. (BDAl)*, Jul. 2022, pp. 51–56, doi: [10.1109/BDAl56143.2022.9862656](https://doi.org/10.1109/BDAl56143.2022.9862656).
- [49] N. Cao and Y. Luo, "Application research of monitoring abnormal behavior analysis and detection method based on GANomaly network," in *Proc. IEEE 2nd Int. Conf. Data Sci. Comput. Appl. (ICDSCA)*, Oct. 2022, pp. 364–369.
- [50] C. Park, S. Lim, D. Cha, and J. Jeong, "Fv-AD: F-AnoGAN based anomaly detection in chromate process for smart manufacturing," *Appl. Sci.*, vol. 12, no. 15, p. 7549, Jul. 2022, doi: [10.3390/app12157549](https://doi.org/10.3390/app12157549).
- [51] O. Sliti, M. Devanne, S. Kohler, N. Samet, J. Weber, and C. Cudel, "F-AnoGAN for non-destructive testing in industrial anomaly detection," in *Proc. 16th Int. Conf. Quality Control Artif. Vis.*, Jul. 2023, p. 46, doi: [10.1117/12.3000063](https://doi.org/10.1117/12.3000063).
- [52] L. Yun, J. Wang, L. Yang, H. Bin, F. Zhang, and L. Gao, "Belt defect detection based on f-AnoGAN," in *Proc. 6th Int. Conf. Adv. Electron. Mater., Comput., Softw. Eng. (AEMCSE)*, Aug. 2023, p. 88, doi: [10.1117/12.3004875](https://doi.org/10.1117/12.3004875).
- [53] Y. Koizumi, S. Saito, H. Uematsu, Y. Kawachi, and N. Harada, "Unsupervised detection of anomalous sound based on deep learning and the Neyman–Pearson lemma," *IEEE/ACM Trans. Audio, Speech, Language Process.*, vol. 27, no. 1, pp. 212–224, Jan. 2019, doi: [10.1109/TASLP.2018.2877258](https://doi.org/10.1109/TASLP.2018.2877258).
- [54] M. Antonini, M. Vecchio, F. Antonelli, P. Ducange, and C. Perera, "Smart audio sensors in the Internet of Things edge for anomaly detection," *IEEE Access*, vol. 6, pp. 67594–67610, 2018, doi: [10.1109/ACCESS.2018.2877523](https://doi.org/10.1109/ACCESS.2018.2877523).
- [55] P. Verma and K. Salisbury, "Unsupervised learning of audio perception for robotics applications: Learning to project data to T-SNE/UMAP space," 2020, *arXiv:2002.04076*.
- [56] F. Di Mattia, P. Galeone, M. De Simoni, and E. Ghelfi, "A survey on GANs for anomaly detection," 2019, *arXiv:1906.11632*.
- [57] Y. Xie, H. Cheng, Y. Wang, and L. Ye, "An efficient temporary deepfake location approach based embeddings for partially spoofed audio detection," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Apr. 2024, pp. 966–970, doi: [10.1109/icassp48485.2024.10448196](https://doi.org/10.1109/icassp48485.2024.10448196).
- [58] H. Tak, J.-W. Jung, J. Patino, M. Kamble, M. Todisco, and N. Evans, "End-to-end spectro-temporal graph attention networks for speaker verification anti-spoofing and speech deepfake detection," 2021, *arXiv:2107.12710*.
- [59] G. Lin, W. Luo, D. Luo, and J. Huang, "One-class neural network with directed statistics pooling for spoofing speech detection," *IEEE Trans. Inf. Forensics Security*, vol. 19, pp. 2581–2593, 2024, doi: [10.1109/TIFS.2024.3352429](https://doi.org/10.1109/TIFS.2024.3352429).
- [60] L. Huang and C.-M. Pun, "Self-attention and hybrid features for replay and deep-fake audio detection," 2024, *arXiv:2401.05614*.
- [61] L. Li, T. Lu, X. Ma, M. Yuan, and D. Wan, "Voice deepfake detection using the self-supervised pre-training model Hubert," *Appl. Sci.*, vol. 13, no. 14, p. 8488, Jul. 2023, doi: [10.3390/app13148488](https://doi.org/10.3390/app13148488).
- [62] A. Khan and K. M. Malik, "Securing voice biometrics: One-shot learning approach for audio deepfake detection," in *Proc. IEEE Int. Workshop Inf. Forensics Secur. (WIFS)*, Dec. 2023, pp. 1–6, doi: [10.1109/wifs58808.2023.10374968](https://doi.org/10.1109/wifs58808.2023.10374968).
- [63] R. Liu, J. Zhang, and G. Gao, "Multi-space channel representation learning for mono-to-binaural conversion based audio deepfake detection," *Inf. Fusion*, vol. 105, May 2024, Art. no. 102257, doi: [10.1016/j.inffus.2024.102257](https://doi.org/10.1016/j.inffus.2024.102257).
- [64] C. Sun, S. Jia, S. Hou, and S. Lyu, "AI-synthesized voice detection using neural vocoder artifacts," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2023, pp. 904–912, doi: [10.1109/cvprw59228.2023.00097](https://doi.org/10.1109/cvprw59228.2023.00097).
- [65] K. W. Cattermole, "The Fourier transform and its applications," *Electron. Power*, vol. 11, no. 10, pp. 35–39, Oct. 1965, doi: [10.1049/ep.1965.0268](https://doi.org/10.1049/ep.1965.0268).
- [66] J. J. Mathew, R. Ahsan, S. Furukawa, J. G. K. Kumar, H. Pallan, A. Singh Padda, S. Adamski, M. Reddiboina, and A. Pankajakshan, "Towards the development of a real-time deepfake audio detection system in communication platforms," 2024, *arXiv:2403.11778*.
- [67] A. Qais, A. Rastogi, A. Saxena, A. Rana, and D. Sinha, "Deepfake audio detection with neural networks using audio features," in *Proc. Int. Conf. Intell. Controller Comput. Smart Power (ICICCSPP)*, Jul. 2022, pp. 1–6, doi: [10.1109/ICICCSPP53532.2022.9862519](https://doi.org/10.1109/ICICCSPP53532.2022.9862519).
- [68] Y.-A. Jung and D.-H. Kim, "Performance analysis by kernel size and pooling layer in MFCC-based environmental sound classification CNN," *J. Digit. Contents Soc.*, vol. 23, no. 5, pp. 913–920, May 2022, doi: [10.9728/dcs.2022.23.5.913](https://doi.org/10.9728/dcs.2022.23.5.913).
- [69] B. McFee et al., "librosa/librosa: 0.10.2," May 2024, doi: [10.5281/zenodo.4923181](https://doi.org/10.5281/zenodo.4923181).
- [70] S. Han, "A study on the development of deep learning-based deep voice detection system using mel-spectrogram and MFCC," *Trans. Korean Inst. Electr. Eng.*, vol. 72, no. 3, pp. 186–192, 2023.
- [71] V. Fernandes, L. Mascareñas, C. Mendonça, A. Johnson, and R. Mishra, "Speech emotion recognition using mel frequency cepstral coefficient and SVM classifier," in *Proc. Int. Conf. Syst. Model. Advancement Res. Trends (SMART)*, Nov. 2018, pp. 200–204, doi: [10.1109/SYS-MART.2018.8746939](https://doi.org/10.1109/SYS-MART.2018.8746939).
- [72] Z. Kh. Abdul and A. K. Al-Talabani, "Mel frequency cepstral coefficient and its applications: A review," *IEEE Access*, vol. 10, pp. 122136–122158, 2022, doi: [10.1109/ACCESS.2022.3223444](https://doi.org/10.1109/ACCESS.2022.3223444).
- [73] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, "Generative adversarial networks," *Commun. ACM*, vol. 63, no. 11, pp. 139–144, Oct. 2020, doi: [10.1145/3422622](https://doi.org/10.1145/3422622).
- [74] Q. Wu, Y. Chen, and J. Meng, "DCGAN-based data augmentation for tomato leaf disease identification," *IEEE Access*, vol. 8, pp. 98716–98728, 2020, doi: [10.1109/ACCESS.2020.2997001](https://doi.org/10.1109/ACCESS.2020.2997001).



DAEUN SONG is currently pursuing the bachelor's degree in information security and data science. She is currently a Passionate Researcher in the field of AI ethics and security. She has experience in machine learning, deep learning, and reinforcement learning. She is also focusing on AI security, including privacy protection and ethical applications of vision technologies. In this project, she contributed to research and idea generation and modeling and experimentation. She hopes that this research will positively impact the detection of deep-fake audio and contribute to the development of more secure and trustworthy AI systems.



Nayoung Lee is currently pursuing the bachelor's degree in information security with Seoul Women's University. Her primary research interests include artificial intelligence, information security, cloud security, and network security. She is currently focusing on developing security solutions utilizing artificial intelligence technologies.



JIWON KIM is currently pursuing the bachelor's degree in information security with Seoul Women's University. Her research interests include artificial intelligence, cybersecurity, and software development.



EUNJUNG CHOI received the B.Sc., M.S., and Ph.D. degrees in computer engineering from Seoul Women's University, in 1997, 2000, and 2005, respectively. She has been an Associate Professor with the Department of Information Security, Seoul Women's University, since 2006. Her research interests include deepfake generative AI and AI for cybersecurity.

...